

1st

UW PICO 5.09

File: Practical5b.c

Modified

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2]; // Pipe file descriptors: fd[0] = read end, fd[1] = write end

    pipe(fd); // Create an anonymous pipe

    // First Child: Runs "ls -l"
    if (fork() == 0)
    {
        dup2(fd[1], STDOUT_FILENO); // Redirect stdout to pipe write-end
        close(fd[0]);                // Close unused read end
        close(fd[1]);                // Close original write end descriptor
        execlp("ls", "ls", "-l", NULL); // Execute "ls -l"
    }

    // Second Child: Runs "grep .c"
    if (fork() == 0)
    {
        dup2(fd[0], STDIN_FILENO); // Redirect stdin to pipe read-end
        close(fd[0]);              // Close original read end descriptor
        close(fd[1]);              // Close unused write end
        execlp("grep", "grep", ".c", NULL); // Execute "grep .c"
    }

    // Parent Process: Close pipe ends and wait for children
    close(fd[0]);
    close(fd[1]);
    wait(NULL);
    wait(NULL);

    return 0;
}
```

⌘ Get Help

⌘ Exit

⌘ WriteOut

⌘ Justify

⌘ Read File

⌘ Where is

⌘ Prev Pg

⌘ Next Pg

⌘ Cut Text

⌘ UnCut Text

⌘ Cur Pos

⌘ To Spell

```
nedikondarakesh@Medikondas-MacBook-Air ~ % nano Practical5b.c
nedikondarakesh@Medikondas-MacBook-Air ~ % gcc Practical5b.c -o Practical5b
nedikondarakesh@Medikondas-MacBook-Air ~ % ./Practical5b
a.c.save
append_st.c
append_stder.c
append_stderr.c
Applications
ar.c
background_jobs
background_jobs.c
background_jobs.dSYM
benchmark_results.csv
cd_dispatch
cd_dispatch.c
cd_dispatch.dSYM
childprocess
childprocess.c
clean_parser.c.save
combined_complex
combined_complex.c
combined_complex.dSYM
custom_shell_parser.c.save
Documents
a.c
error_summary.txt[cite: 7]
executor
executor.c
executor.dSYM
export_state.c
file_proc
file.c
fork_demo.c
history_pipeline.c
integrated_shell.c
io_redirection
io_redirection.c
io_redirection.dSYM
job_listing_fg.c
Main.class
memory_and_test_shell.c
```

```
history_pipeline.c
integrated_shell.c
io_redirection
io_redirection.c
io_redirection.dSYM
job_listing_fg.c
Main.class
memory_and_test_shell.c
memory_test.c
multi_fork.c
multi_pipe.c
multi.c
Music
null.c
OrbStack
ossp.c
output_users.txt[cite: 7]
p.c
parentprocess
parentprocess.c
parser_test.c
parser_test.c.save
parser.c
path_waitpid.c
Pictures
pipe.c
pipeline_perf_shell.c
Practical5b
Practical5b.c
project
Project.c
Public
q.c
quotes_parser.c
r.c
shell_complex
shell_complex.c
shell_complex.dSYM
shell_history.c
shell_parser.c
signal_handler.c
system.c
t.c
task4.c
vars_dispatch
vars_dispatch.c
vars_dispatch.dSYM
w.c
medikondarakesh@Medikondas-MacBook-Air ~ %
```

2nd